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PROJECT AND TEAM INFORMATION

Project Title

OS User Activity Monitoring & Insider Threat Detection System

Team information	Details
Team name and number	KernelGuard OSDBMS-V-2026-T298
Mentor	(Prof.)Dr.Mohammad Wazid
Team member 1 (Team Lead)	Arshpreet Singh 24021835 24021835@geu.ac.in 
Team member 2	Mohd Ahmed Khan 24021365 24021365@geu.ac.in 
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PHASE-III PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

KernelGuard Phase III combines Linux Audit events with sampled user, session, process and USB information. Five configured rules create alerts with event evidence and a saved policy snapshot. An authenticated dashboard supports alert review and USB approval decisions while retaining earlier actions. The system uses relational transactions so events, inventory changes and their related records remain consistent. It runs as a SQLite demonstration and in separate MariaDB and Oracle MySQL 8.4 Kali deployments.

Updated Project Approach and Architecture (2 pts)

The audit collector groups syscall and authentication records; inventory polling reads account, process and USB information from Linux facilities. Process identity includes host, boot ID, PID and start time to avoid confusing reused PIDs. Incomplete scans retain warnings instead of treating unseen processes as exited. A writer lock coordinates transactions for events, alerts, checkpoints and inventory transitions. The relational schema links alerts to supporting events, policy settings and review history. Flask sessions, CSRF checks and stale-decision checks protect administrator actions. The five rules remain deterministic and explainable rather than using a trained model.

Tasks Completed (7 pts)

Task Completed	Team Member
Audit collection and host-data integration	Arshpreet Singh
Persistence, schema and transactional joins	Mohd Ahmed Khan
Five policies and evaluation scenarios	Ankit
Dashboard, authentication and review workflow	Mohd Shoaib

Challenges/Roadblocks (7 pts)

Boot changes and PID reuse require stronger identity keys than PID alone. Audit restart replay must not create duplicate events. Process scans can be partial because of permissions or exit races, so the application records completeness and warnings. A new USB device can match a rule without being malicious, and an authorized backup can meet the bulk-file threshold. The alert therefore carries evidence for human review. Physical USB remove/reconnect and live enumeration-failure checks remain open on the available lab hardware.

Project Outcome/Deliverables (2 pts)

The deliverable is a working local host-monitoring prototype with five policy families, user/process/USB inventory, linked evidence and administrator decision history. A separate Oracle MySQL 8.4.11 Kali deployment runs alongside the existing MariaDB deployment. Saved evaluation data contains 11 of 11 expected synthetic policy outcomes. The software flags configured policy matches; it does not stop activity or prove malicious intent.

Progress Overview (2 pts)

Core implementation, demonstration data, automated regression tests and documented Kali audit/inventory checks are complete. A saved SQLite workload processed 500 synthetic events with 100 alerts in 0.4806 seconds; it excludes collection and web rendering. The available Kali host has no spare removable device for physical removal/reconnection, enrollment and revocation checks. Live USB-enumeration failure handling also remains to be exercised. No field precision or recall is claimed.

Codebase Information (2 pts)

Local source: C:\KernalGuard. Branch: main. The visible repository has a project snapshot and MySQL deployment documentation commits, plus current working-tree changes. Main modules: kernelguard/core.py, activity_rules.py, parser.py, inventory/, auth.py, reviews.py and web.py. Evidence documents: docs/ARCHITECTURE.md, docs/EVALUATION.md, docs/ACCEPTANCE.md and docs/MYSQL84_DOCKER.md. No public remote is configured in this copy.

Testing and Validation Status (2 pts)

Test Type	Status	Notes
Local automated tests	89 passed, 1 skipped	Current Windows checkout
Policy evaluation	11/11 expected	Synthetic, hand-authored cases
Oracle MySQL 8.4	Pass	Recorded Kali integration run

Deliverables Progress (2 pts)

Completed: five policies, evidence-linked alerts, host and USB inventory, admin review, SQL schema, setup notes and regression tests. Recorded live checks cover audit, inventory and review on Kali. Pending hardware acceptance: physical USB removal/reconnection and live enumeration-failure behavior.